

Bismillah Khan

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Education

B.Tech in Computer Science Engineering, IIIT Bhubaneswar
CGPA: 8.02

2023 – 2027

Professional Summary

AI/ML Engineer with hands-on experience building modular, scalable systems and high-accuracy predictive models. Proven track record in developing an LLM-driven voice assistant with optimized vector-based memory and a spam detection engine achieving 98.2% accuracy and 1.0 precision. Proficient in Python, C++, and modern AI frameworks (LangChain, Hugging Face), dedicated to delivering high-performance, user-centric AI solutions.

Projects

Friday AI — Intelligent Voice Assistant

[GitHub Link](#)

Tech Stack: Python · LangChain · Hugging Face Transformers · Gemini API · FAISS · SentenceTransformers · gTTS · pyttsx3 · Spotify/YouTube APIs

- Architected a **modular, LLM-driven voice assistant** using **LangChain** (Chains, Agents, Prompt Templates) for task orchestration, dynamic API routing, and adaptive personality-driven interactions across 3 personality modes (friendly, professional, technical); achieved **end-to-end response latency of 1.4s** via async API calls and pre-compiled prompt templates.
- Engineered a **semantic long-term memory system** using **FAISS** and **SentenceTransformers** vector embeddings to replace linear history scanning; approximate nearest-neighbour search over **500+ stored interactions** reduced memory retrieval latency to **under 80ms**, enabling scalable cross-session recall without degrading response time.
- Designed a **two-pass speech preprocessing pipeline** — combining a custom **filler-word filter** (“uh”, “uhh”, “hmm”, “you know”) with **NLTK stopwords removal** — to sanitize raw voice transcriptions before LLM dispatch; eliminated noise from **37% of real-world voice inputs** and improved **intent recognition accuracy from 71% to 91.4%** on a 200-sample test set.

Spam Email Classifier — Chrome Extension

[GitHub Link](#)

Tech Stack: Python · Flask · NLTK · Scikit-learn · Pandas · TF-IDF · Random Forest · SVM · Naive Bayes · Chrome Extension

- Engineered an **end-to-end NLP pipeline** for real-time spam detection, including text preprocessing (tokenization, stopwords removal), **TF-IDF feature extraction**, and model inference via a lightweight **Flask REST API**.
- Designed and benchmarked multiple **machine learning classifiers** (Random Forest, SVM, Naive Bayes, Extra Trees); implemented a Soft Voting Ensemble, achieving **98.2% accuracy** and a **precision score of 1.0** on the test set.
- Developed a **Chrome browser extension** as the client interface, enabling real-time email classification with sub-second inference latency via seamless backend integration.

Technical Skills

Languages & Tools

Python, C++, MySQL, FastAPI, NumPy, Pandas, Git

Frameworks & Libraries

LangChain, Hugging Face Transformers, Scikit-learn, FAISS, SentenceTransformers, gTTS, Pytorch

Techniques

Generative AI, Large Language Models (LLMs), NLP Pipelines, Time Series Forecasting, Machine Learning, Agentic AI, Retrieval-Augmented Generation (RAG)

Interests

Data Structures & Algorithms, AI/ML, Creative Project Development